Final Project Proposal

a) The largescale problem we are trying to solve is the dependence on fossil fuels and environmental impacts of hydropower plants. The problem we are trying to solve with our project is our lack of power to our devices such as a pump and lights.

b) The criteria we set for our project:

Absolute thresholds:

- Our generator needs to produce a minimum of 6 watts to power the pump at 4.2 watts and LED lights at a 1 watt maximum
- The project needs to be under 5 square feet of surface area to accommodate travel

Priorities

- High efficiency. A typical hydroelectric dam harnesses about 90% of the flowing water's energy. With our materials and methods we hope to shoot for around 70% efficiency.
- Low cost. Since we already own many of the materials, we would like to keep our cost under $50

c) Alternative designs:

- The first design includes a reservoir at a higher altitude. The water would flow through a tube and then through the turbine which would then spin and with the generator would create power. The water would keep flowing into a pond. This pond would then fill up and overflow into a river which would travel and end up in a hidden reservoir. The pump would take the power produced by the generator and pump from the lower reservoir into the higher one. Our design would also include miniature landscape surrounding the dam.
- The second design would incorporate the same design of reservoirs. The higher reservoir will store water that would then flow through the tube and through the turbine. The power generated would power lights throughout a model town. The water would keep flowing to a lower reservoir. There would be a model fish ladder to help show environmental awareness. For this design we would have to keep refilling the top reservoir.
- Our last design idea would incorporate producing power for both the lights and a pump. In this design the power generated would be stored in a battery. When enough power is stored to run the pump, the pump would turn on and repump the water from the lower reservoir to the higher one.
- Throughout all these designs there are a lot of variables that could change based on what performs most efficiently. These include: number of coils/magnets, number of turns in the coils, strength of magnets, stator-rotor distance, wire gauge, turbine design (fan vs. wheel), penstock length, penstock cross-sectional area, penstock pitch, penstock GPM, reservoir size and shape, penstock inlet height, turbine orientation, stator-rotor diameter, capacitor usage, flywheel weight usage, bearings used.

d) When ranking our designs, the most important thing we're looking for is to reach our goal of producing 6 watts. We also need to meet the absolute threshold of 5 sq. feet of surface area. Of the two design priorities, the efficiency is more important than the cost. We will assign weightings of 70% for efficiency and 30% for cost.

e) We plan to test our design with the use of a multimeter to test the output of the generator. We will then convert the output energy into a percentage of potential water energy harnessed. When these numbers match our requirements we will test our project to see if the pump will take the water from the lower reservoir and pump it back into the higher reservoir. If we only decide to power lights, then our generator should produce enough power for our lights to turn on.